

Case Book

HEC MBA Consulting Club

Spring 2018



4 sections to help you hone case interviews skills

1

Introduction

Why case interviews
How to use Case book

2

Trainer

Case preparation
Case facilitation
Structured feedback

3

Trainee

Soft vs. hard skills
Case preparation
Structured learning

4

Practice

Cases
Drills
Fit

1

Introduction



Keeping in mind why case interviews are performed to put you in the right mindset

Why case interviews are used in consulting?

- To assess candidates' skills used every day on the job in management consulting

Which skills are being assessed?

- ✓ Empathy
- ✓ Problem solving
- ✓ Enthusiasm
- ✓ Ownership
- ✓ Structure
- ✓ Executive speak
- ✓ Business acumen
- ✓ Creativity

What case interviews are not?

- ✗ A math exam
- ✗ Memory test on frameworks
- ✗ A speed run
- ✗ A knowledge test

Reading a case book is not a substitute to practicing and structured learning

A case book designed to support your preparation...



How to use case book

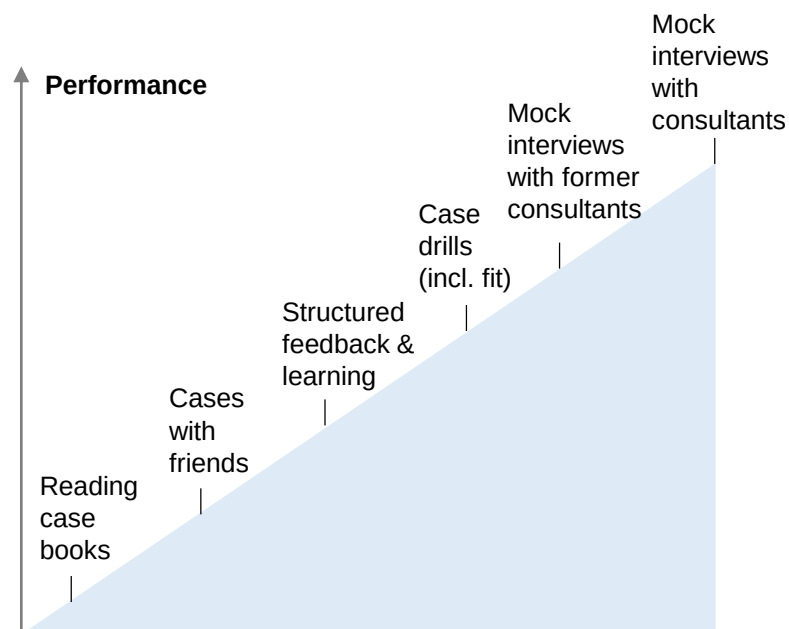
- ☐ Read Introduction, Trainer and Trainee sections to get familiar with case interviews format and expectations
- ☐ Observe a case offered by a friend to another friend (shadowing)
- ☐ Receive cases and ensure structured learning and performance improvement tracking
- ☐ Perform case drills (including fit) to hone skills on specific parts of case interviews
- ☐ Master one case as a trainer and offer it to a few friends
- ☐ Actively track performance of trainees and provide structured feedback

...however, reading it is not enough to ensure interview readiness.



Activities that can help you master case interview skills

Non-exhaustive



2

Trainer



Case trainers should dedicate ~1hr/case for preparation

How to prepare a case as a trainer?

Trainer's preparation checklist

- ☐ Read case thoroughly
- ☐ Get comfortable with case flow
- ☐ Learn main questions and areas to assess
- ☐ Know where to find main figures and stats of the case
- ☐ Try to answer case by yourself

Dedicate at least **~1hr**
to prepare the case to
ensure great session
with trainee

A trainer has the flexibility to calibrate case difficulty

Advice: As a trainer you have the flexibility to adapt the flow of the case session

How to facilitate case

- Arrive before meeting time to prepare case and feedback sheet
- Explain meeting format to trainee before starting
- Give case prompt
- Cover major parts of case based on suggested case flow
- Actively listen to candidate answers and make it interactive
- Calibrate case to candidate's level
- Take notes to provide trainee specific feedback on performance
- Keep track of time – a case should last between 30 and 45 min. followed by 10-15 min. feedback

How to calibrate case difficulty

- Ask candidate 'why do you think this data would be useful' when he/she asks for data
- Ask trainee to explain her/his rationale when making an assumption
- Providing data only if candidate asks for it
- Request for more ideas on how to solve problem (creativity questions)



Structured feedback is key to learning and performance enhancement of trainees

To embrace



How to provide feedback as a trainer

- ✓ Dedicate at least 10 min. to feedback
- ✓ Make it a discussion with the trainee
- ✓ Ask the candidate how he/she thought the case went
- ✓ Provide feedback what the candidate did well
- ✓ Provide feedback on areas that present room for improvement
- ✓ Focus discussion and feedback on skills that can apply in other cases (i.e., transferable skills)
- ✓ Suggest ways for the trainee to practice and improve skills

To avoid



What not to do when providing feedback

- ✗ Don't let the trainee talk
- ✗ Focus too much on case book answers rather than feedback on transferable skills
- ✗ Lecture the trainee
- ✗ Be 'too nice'; trying to please trainee by providing only positive feedback
- ✗ Dedicate less than 5 minutes to feedback
- ✗ Being not specific enough on performance of trainee
- ✗ Forget to give feedback to all trainees (group sessions)

A template to help trainers offer structured feedback

Advice: This evaluation grid is not meant to be exhaustive; be flexible and offer tailored feedback based on the case

Area	Evaluation Criteria Did the candidate...	Score 1 (worst) to 10 (best)
Structure	- Tailor structure to the client's question? - Has at least 2 sublayers in structure? - Develop a structure in 30-45 sec.? - Frame structure as questions? - Develop a MECE structure? - Executive summarize structure before deep diving? - Present structure in ~2 min.?	
Maths	- Walk the trainer through calculations' rationale? - Do calculations properly? - Ask before rounding figures? - Do calculations fast enough? - Stay engaging? - Sanity check results?	
Market Sizing	- Present approach before asking for figures? - Identify all key factors to size the market? - Present 'reasonable' assumptions?	
Creativity	- Show creative judgement? - Embrace client's point of view? - Develop solutions in a structured way?	
Soft Skills	- Present enthusiasm to solve business problem? - Show empathy? 'Carry the monkey' throughout the case? - Show assertiveness, yet no over-confidence?	
Conclusion	- Use a pyramid structure? - Present his/her recommendation in less than a minute? - Executive summarize recommendation? - Present future work and potential risks? - Show some 'selling' skills?	
Comments		Average Score: (out of 10)

Note: Cases might not present all evaluation criteria, trainer's judgement is advised

3

Trainee



Both soft and hard skills are being assessed during case interviews



Soft Skills

What interviewers might be reflecting upon...



- Has the candidate the right profile for the job?
- Would she/he embarrass me in front a client?
- How well can the candidate handle pressure?
- How clear the candidate can communicate?
- Is the candidate 'carrying the monkey' and assuming ownership of the problem?
- Is the candidate genuinely interested in the problem?
- Is the candidate listening?
- How would the candidate face an executive?
- Would I spend 6hrs in an airport with him/her?

...and what candidates can do to shine during interviews (not exhaustive)

- Listen
- Engage with interviewer
- Focus on end goal, not on mechanisms of cases
- Be genuinely interest in the client's problem
- Show empathy
- Own the problem; don't try to it pass it on to the interviewer
- Don't 'disappear'; make it conversational

Hard Skills

- How structured is the candidate?
- Can the candidate find 'so whats' (i.e., an insight relevant to the case) when presented a chart?
- How comfortable is the candidate with playing with numbers?
- Can she/he show good business acumen?
- How does the candidate react to new information?
- How creative and structured is the candidate when problem solving?

- Loose preconceived assumptions; validate assumptions with interviewers
- State insights/'so whats' when receiving data
- Benchmark data to identify insights
- Use a Mutually Exclusive, Collectively Exhaustive (MECE) structure to solve business question
- Present structure, yet creative thinking
- Be comfortable with basic math problems (no calculator allowed)

Practice is key for successful interviews

Advice: Use RocketBlocks.me, PrepLounge.com and other online resources to hone your skills

To embrace



How to prepare for consulting interviews

- ✓ Practice case drills
- ✓ Prepare & rehearse answers to fit questions – key for successful interviews in consulting¹
- ✓ Train in environment close to real interviews setting
- ✓ Train with consultants and former consultants
- ✓ Do cases with multiple partners to learn different interviewing styles
- ✓ Develop pre-performance routine for a clear mind (e.g., taking a deep breathe, repeating a cue words, rehearsing a sequence of movement²) to reduce stress and enhance performance
- ✓ Read FT, The Economist and other business publications to develop basic business knowledge
- ✓ Ask yourself business questions about the world around you (e.g., I wonder how this coffee shop makes money?)

To avoid



What to avoid in case preparation

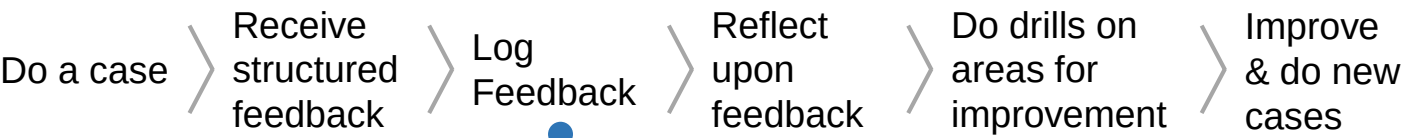
- ✗ Spending too much time in passive training
- ✗ Allocating not enough time to fit questions practice
- ✗ Using buzzwords (e.g., frameworks, clarifying questions, “this is a market sizing question”)
- ✗ Getting attached to canned frameworks (e.g., 4Ps, Porter’s 5 forces, 3Cs, etc.)
- ✗ Focusing too much on case solutions instead of developing case skills
- ✗ Being too rigid and not tailoring approach to interviewer

¹<https://www.cnn.com/2018/06/06/how-to-land-a-job-at-mckinsey.html>

²https://youtu.be/CqgmzFr_GM

Creating a case practice log will help targeting areas for improvement

Getting better at Case Interviews



A suggested template to log and track your progress

Case Practice log
Tracking Your Progress

#	Date	Case	Case type	Source	Trainer	Performance (out of 10)	Feedback	
							What did I Do Great?	What Do I Have To Improve For Next Cases?
1	October 4th 2018	SunFurry	Market Entry / Value Sizing	Haas Casebook	John Smith	8	- Used a MECE structure - Shown interest for solving business problem	- Make sure I understand the prompt before structuring my thoughts on how to tackle the case - Engage with the trainer while performing math calculations
2								
3								
4								
5								
...								

4

Practice



Case solutions are indicative; flexibility & creativity are advised

Case difficulty
Found on 1st page of cases

- EASY
- MEDIUM
- HARD

Before starting...

- A case is first and foremost a situation where you're being asked to help someone with a business problem
- Your enthusiasm and creativity are welcomed as business solutions can be solved by embracing multiple solutions
- Therefore, case solutions found in this case book are presented as indicative
- Data is fictional and should not be considered as representative of reality

Happy
training!

Downstream Oil

Trainer's guide

- TO PROMPT
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- SUGGESTED SOLUTIONS

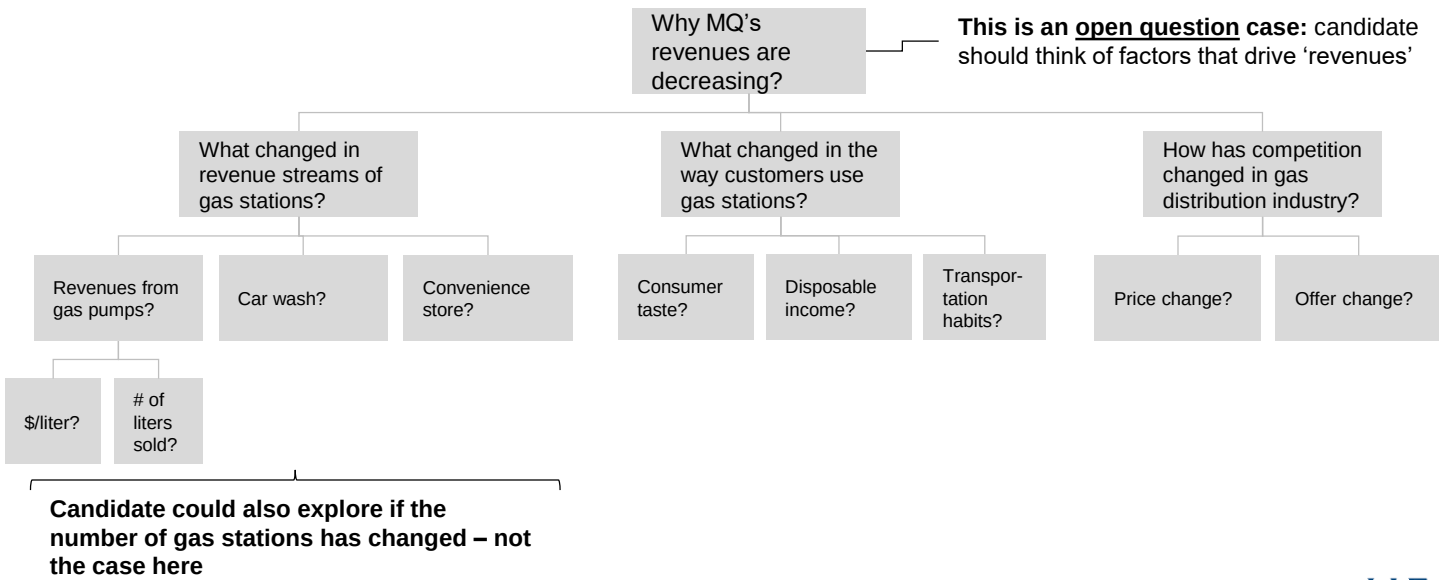
Opening statement

Your client is in the business of oil distribution. The company, MQ, operates in multiple locations across the United States, from Vermont to Hawaii. MQ has been profitable for many decades which allowed it to offer generous employee compensation and benefits. However, tide is now shifting: the company has seen its revenue stumble for the first time last year, a situation that its managers have never experienced before. Hence, the Sales SVP has reached out to you and your team to address the issue. The client is preparing her next steering committee where the situation will be discussed and would like your help to tackle two questions: **Why revenues from their core business are decreasing? How can they improve the situation?**

Further information

- MQ's core business: Gas stations
- 1500 locations
- Products: gas, car wash, snacks and soft drinks (i.e., through convenience store attached to gas stations)
- Sells directly to consumer
- Only operates gas stations (i.e., value chain not integrated)
- Located across the US but mainly in cities and metropolitan areas
- Typical gas station: gas pumps and convenience stores
- No specific target regarding revenue increase
- Employees are all permanent (i.e., no contractors)

Suggested structure



Downstream Oil

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Market sizing question

The client agrees with your structured approach on how to tackle the issue, but would like a better idea on how big is the current market for gas stations in the US, excluding revenues from snacks, drinks and car wash. **How would you help her address this question?**

Further information

- US population: 330M
 - Average household: 3 people
 - Average # of cars: 1/household
 - Average tank capacity/car
 - Compact: 50 liters
 - SUV: 100 liters
 - For simplicity, no other type of cars on market
 - SUVs: 50% of market
 - On average a car does 1000Km/2 weeks
- Efficiency
 - Compact: 1 000km/tank
 - SUV: 500km/tank
 - Type of gas used + Price
 - Compact: Octane 87 sold at \$0.5/liter
 - SUV: Supreme sold at \$1/liter
 - No difference in efficiency between cars riding in metropolitan areas and countryside
 - Demand from trucks/boats is excluded
 - Our client has a market share of 30%
- Bonus points if candidate ask about these variables to refine her/his estimates

Suggested approach

- Top-down approach**
- US pop: 330M
 - Avg. HH: 3 people
 - # of HH: 110M
 - Avg. # of cars/HH: 1
 - # of cars in US: 110M
 - Divided in 2 segments: Compacts (50%) and SUVs (50%)
 - 55M Compacts & 55M SUVs
 - Compacts:
 - Price/liter: \$0.5
 - Avg. tank small car: 50 liters
 - Avg. consumption: 1000Km/2 weeks (50 weeks/yr. is OK)
 - Efficiency: 1000km/tank
 - SUV:
 - Price/liter: \$1
 - Avg. tank: 100 liters
 - Avg. consumption: 1000Km/2 weeks (50 weeks/yr. is OK)
 - Efficiency: 500km/tank

\$309B/yr.
(\$34 375M - Compacts
\$275 000M - SUV)

Most important is not the final answer but candidate's approach to define size of market

Bonus question

What's the total revenue of our client if it has a 30% market share?

Answer

\$92.7B

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Chart question

The size of the market of gas in the US is indeed impressive. Your client knows that gas stations have also other ways to generate revenues and has just received some historical data and projections on their different revenue streams. She provided you with the following chart (present next page's chart to trainee). **Based on what you see here, what can you conclude?**

Further information

- Car wash revenues peaked in 2016 as a national study showed that good maintenance of car could increase its resale value
- Convenience store revenues in absolute terms are not growing - they are quite stable
- The retail price of gas is decreasing from 2012 and on. However, the demand for gas in volume is stable
- Footnote: other revenue source = electric charging service (provide 2nd chart if requested)
- Few players are acting on the electric charging market
- Gas stations are operating in a highly fragmented and commoditized market

Insights from chart

1st chart

- Convenience store: largest part of revenues (2/3rds)
- Gas: smallest part of revenues. However, this product is acting as a loss leader and is important for the business (i.e., customers coming for gas but also buying candies, soft drinks, etc.)
- Footnote: other revenues are not to neglect
- Revenues are in relative figures and not in absolute terms; be careful when interpreting

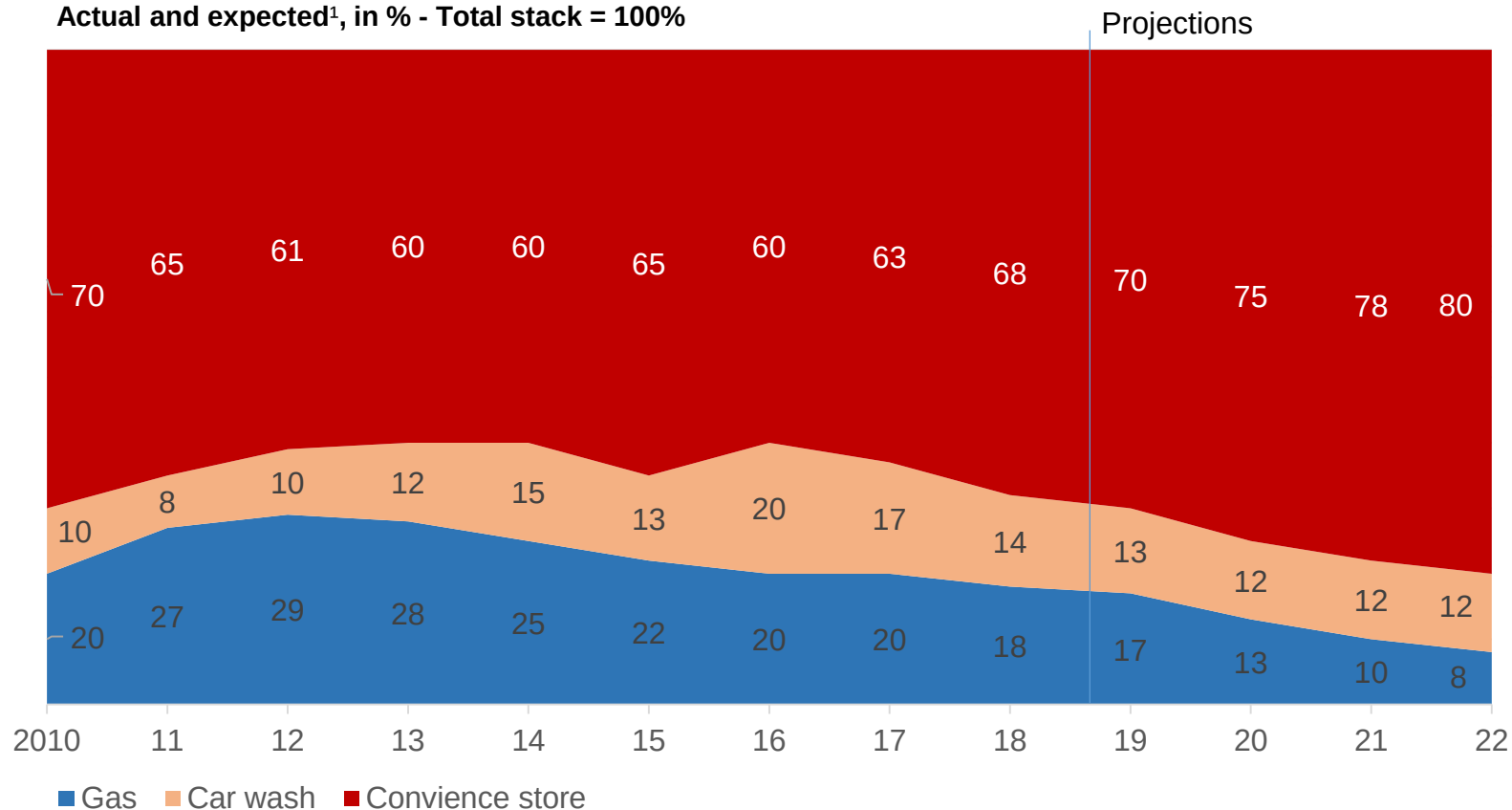
2nd chart

- Electric charging is a new revenue stream potential
- However, implementing electric charges in current gas stations might require new capabilities

Downstream Oil

Gas stations revenue streams

Actual and expected¹, in % - Total stack = 100%



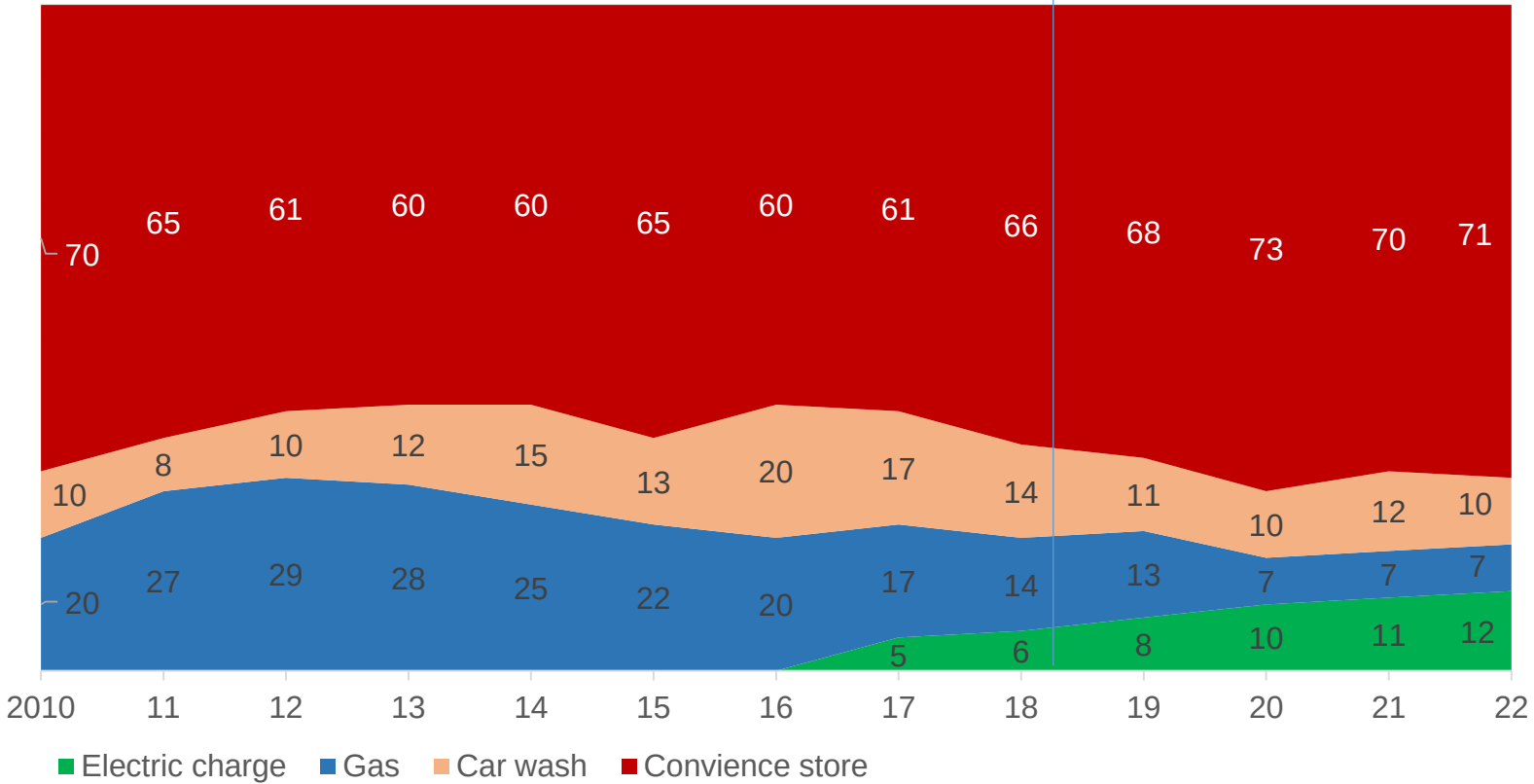
¹Total revenue is expected to grow from other revenue sources not represented

Downstream Oil

Gas stations revenue streams – Including other revenue sources

Actual and expected, in % - Total stack = 100%

Projections



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Creativity question

Additionally, your client has learned from a recent survey that customer satisfaction towards gas stations is shrinking rapidly. **How would you improve customer satisfaction in this business?**

Suggested answered

- Lower prices
- Faster service
 - Reduction of bottlenecks and wait lines at gas stations
- Full-service: no need to get out of car
- 'Uber-like' service for gas
- At home gas delivery
- Etc.

Creativity is expected here. In order to get points here, trainee has to provide answers that are creative and embracing customer's point of view

Conclusion

The SVP of Sales is meeting with the SteerCo in 3 minutes and would like your brief on what to do to tackle the issue. **It light of what you have learnt, what do you recommend?**

Suggested answered

- This case is open ended (no precise answer suggested)
- Candidate's recommendation has to be structured and based on factual data provided in the case

Bonus points for creative solutions

Consultancy

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Opening statement

Elizabeth & Co, generally known as “Eli & Co”, is a major consultancy operating worldwide. Recently, the company has been facing increasing competition from players such as Accenture, the Big 4 and Roland Berger going directly after its core business. In order to preserve its dominant position in the consulting industry, partners of Elizabeth & Co are wondering if they should acquire another firm. More precisely, the company is contemplating the takeover of PSG Consulting. Therefore, your team has been called in to address this pressing issue: **should Eli & Co acquire PSG consulting?**

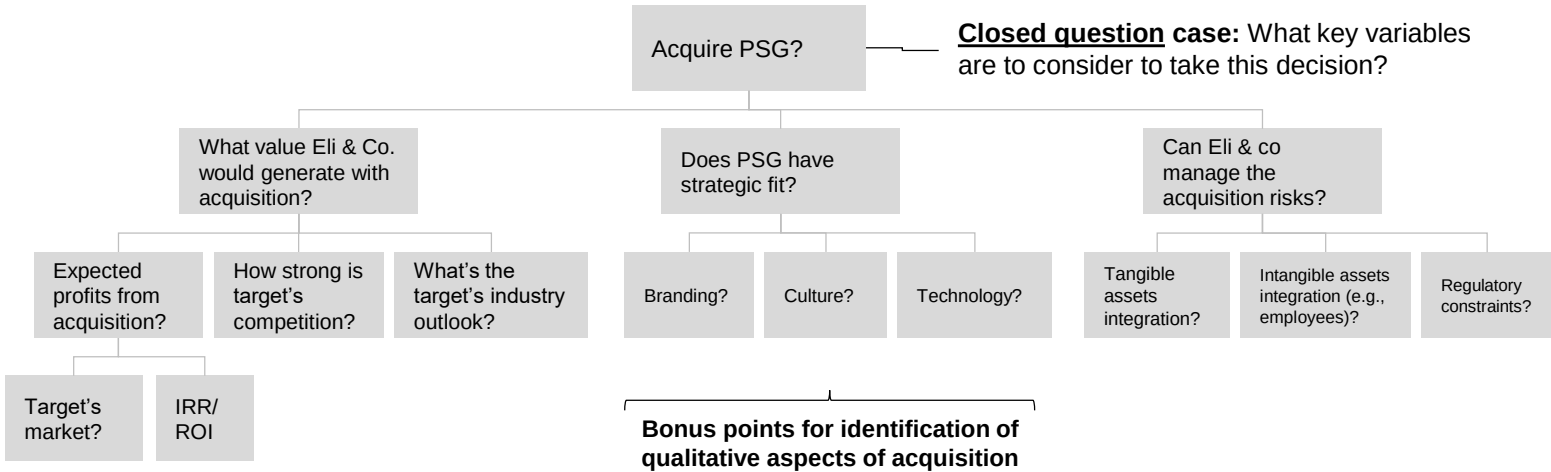
Further information

- Eli & Co**

 - Core business: Strategy consulting (100% of its revenues)
 - Weighted Average Cost of Capital (WACC): 15%
 - Specialized in non-IT consulting
 - Rely mostly on 2 major clients in the banking sector (90% of revenues come from these clients)
 - 1 000 employees
 - Culture: Achievement oriented
 - Operating worldwide
 - Limited partnership ownership structure
- PSG Consulting**

 - Core business: Implementation of business transformations – representing 66% of topline
 - Growing business
 - Specialized in non-IT consulting, but has IT capabilities
 - 300 employees
 - Multiple clients
 - Culture: focused on client satisfaction
 - Operating in Germany only
 - Publicly traded company

Suggested structure



Consultancy

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Valuation question

Interesting, now let's take a look at PSG's financial position. Solely based on a financial perspective, what would be the value of the company?

Suggested insights

- B/S**
 - Company is highly leveraged
 - High accounts receivables (current payment terms for clients are extremely generous)
 - Has a patent pending on a innovative transformation implementation approach
- P&L**
 - Profitable business
 - High cost of services sold - 70% of revenues (i.e., high salaries of consultants)
 - Topline is growing 10%/yr.
- CFS**
 - Company is generating good free cash flows
 - Debt is weighting high in cash outflows
- Value of PSG Consulting: \$260-280M**
 - 10% growth rate – can be assumed to growth at this rate for the foreseeable future
 - WACC: 15%
 - Eli & Co wants to use a 'football field valuation' (i.e., defining a valuation range by combining methods) to value PSG:
 - (1) Actualized FCF method
 - Current FCF: \$14M
 - Consider the company will last and grow forever (perpetuity)
 - $14M / (15\% - 10\%) = \$280M$
 - (2) Comparables method (comps)
 - Comps trading at 14.4x EBITDA
 - Value = $18M * 14.4 = \sim \$260M$
 - PSG is trading at a discount
 - Alternative (and acceptable) solution:
 - $EV = \text{Market value of Equity} + \text{Net Debt}$
 $(180M + 74M - 16M = \$238M)$

Chart question

The client has just shared with us some insights on PSG (prompt staked column chart to trainee). What can you conclude from this chart?

Suggested insights

- Target company offers a potential for diversification
 - Multi-sectors
 - Not concentrated in financial services as Eli & Co
- Bonus points if trainee identifies potential integration risks related to diversification

Consultancy

PSG Consulting Financial Statements

Simplified Balance Sheet

All figures in \$M	2018A
Assets	
Cash and balances	16
Accounts receivables	98
Other assets	18
Fixed assets	15
Total Assets	147
Liabilities & Equity	
Accounts payables	66
Long term debt	74
Equity	6
Retained earnings	1
Total Liabilities & Equity	147

Simplified P&L

All figures in \$M	2018A	2019E
Revenues	90	99
Cost of good sold	63	69.3
SG&A	9	9.9
EBITDA	18	19.8
EBIT	17	18.9
Interest	4	3
Taxes	1	1
Net income	12	14.9

Simplified CFS

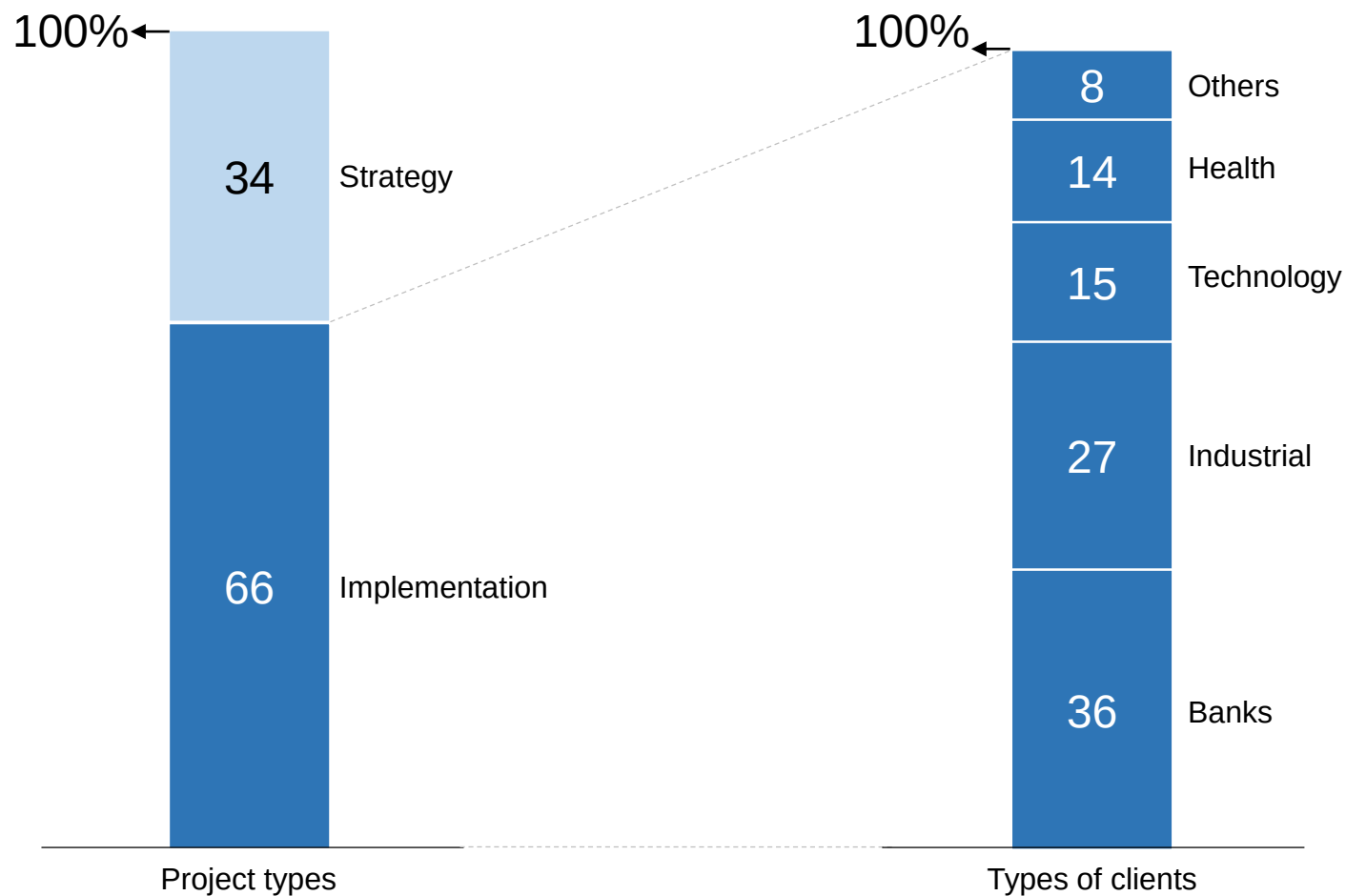
All figures in \$M	2018A
Cash from operating	18
Cash from investing	(4)
Free cash flow	14
Cash from financing	(10)
Change in cash	4
Closing cash balance	16

Note 1: Other assets include patents pending on proprietary implementation approach

Note 2: Current market value of equity = \$180M

Consultancy

PSG Consulting's revenues detailed
In % of revenues



Consultancy

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- SUGGESTED SOLUTIONS

Conclusion

So, what do you think Eli & Co should do?

- “I addressed the following question: Should Elizabeth & Co buy PSG Consulting?”
- My recommendation is: buy the company
- For 3 reasons
 - Financial: Revenues are growing at a 10% rate and present profitability
 - Diversification: Acquiring PSG would allow your company to diversify your client base and unlock cross-selling opportunities (e.g., selling strategy and transformation implementation projects to the same client)
 - Fit: PSG's culture is ‘client centered’ and Eli & Co defines itself as ‘achievement oriented’. Both cultures are compatible
- However, there are some risks to mitigate if the client proceeds with the acquisition
 - Target company is publicly traded and Eli & Co might have to pay a high premium to acquire PSG
 - PSG is highly leveraged and financing might be an issue
 - Company is trading at a high multiple of EBITDA, be careful of overvaluation
 - Integration might be difficult as the PSG's activities are much more diversified than Eli's”

Indicative answer.
Bonus points if
trainee combines
both quantitative
and qualitative
rationale for or
against the
acquisition

Suggested answered

Trainer's guide

TO PROMPT

TO PROVIDE IF REQUESTED

SUGGESTED SOLUTIONS

Opening statement

Benedict, a thriving clothing retailer in Canada is pondering international expansion. “Where to go next?” is often heard in executive talks. However, these discussions usually comes with tactical concerns on how would to go about the expansion. More precisely, Benedict’s management team is puzzled about distribution questions such as if a new market entry should be realized through online, brick and mortar stores or both distribution networks. **You just have been mandated by Benedict’s managers to bring clarity to these questions.**

Further information

Benedict’s situation

- Strong brand equity locally but not internationally
 - Focused on long term growth
 - Privately owned (i.e., not a publicly traded company)
 - Management is typically risk averse to large investments
 - Has a network of stores and an online shop
- Currently 60% of sales come from retail and 40% online
 - Company offers clothes for men and women
 - Revenue distribution: 66% women, 34% men
 - Women spend 3x as much on clothes than men do in Canada

Suggested structure

Option selection case: a matrix structure can help compare options among them and choose best one

	What would generate more growth?	What has the highest implementation cost?	Where do customer prefer to shop?	What would allow for competitive advantage on market?	What is closer to the company's core capabilities?
Retail					
Online					
Both					

Note: at this point, no information is shared on countries that the company is considering for international expansion

Clothing retailer

Trainer's guide

TO PROMPT

TO PROVIDE IF REQUESTED

SUGGESTED SOLUTIONS

Chart analysis

Prompt 1st chart to trainee. **Based on this chart, what can you conclude?**

If trainee has interest in learning more about 'high growth' markets, prompt 2nd chart.

Further information

- Australian market is highly fragmented with high number of low-cost retail players in the clothing industry
- US presents slow growth even if a large market
- American, Chinese & Russian governments are contemplating imposing tariff to foreign companies operating in their market
- In Chile, mostly men shop for clothes (66% of sales)
- In Australia, about 75% of revenues of clothing companies come from women
- Brazilian market presents high degree of political uncertainty

Suggested insights

Chart 1

- Market is divided in 4 types of countries:
 - Booming (+10% growth):
 - Russia, China, Brazil, Chile and Australia
 - Solid growth (5-10% growth):
 - Canada, France, Germany, Sweden and Norway
 - Slow growth:
 - Saudi Arabia,, India, US and Finland
 - Undocumented:
 - All other countries

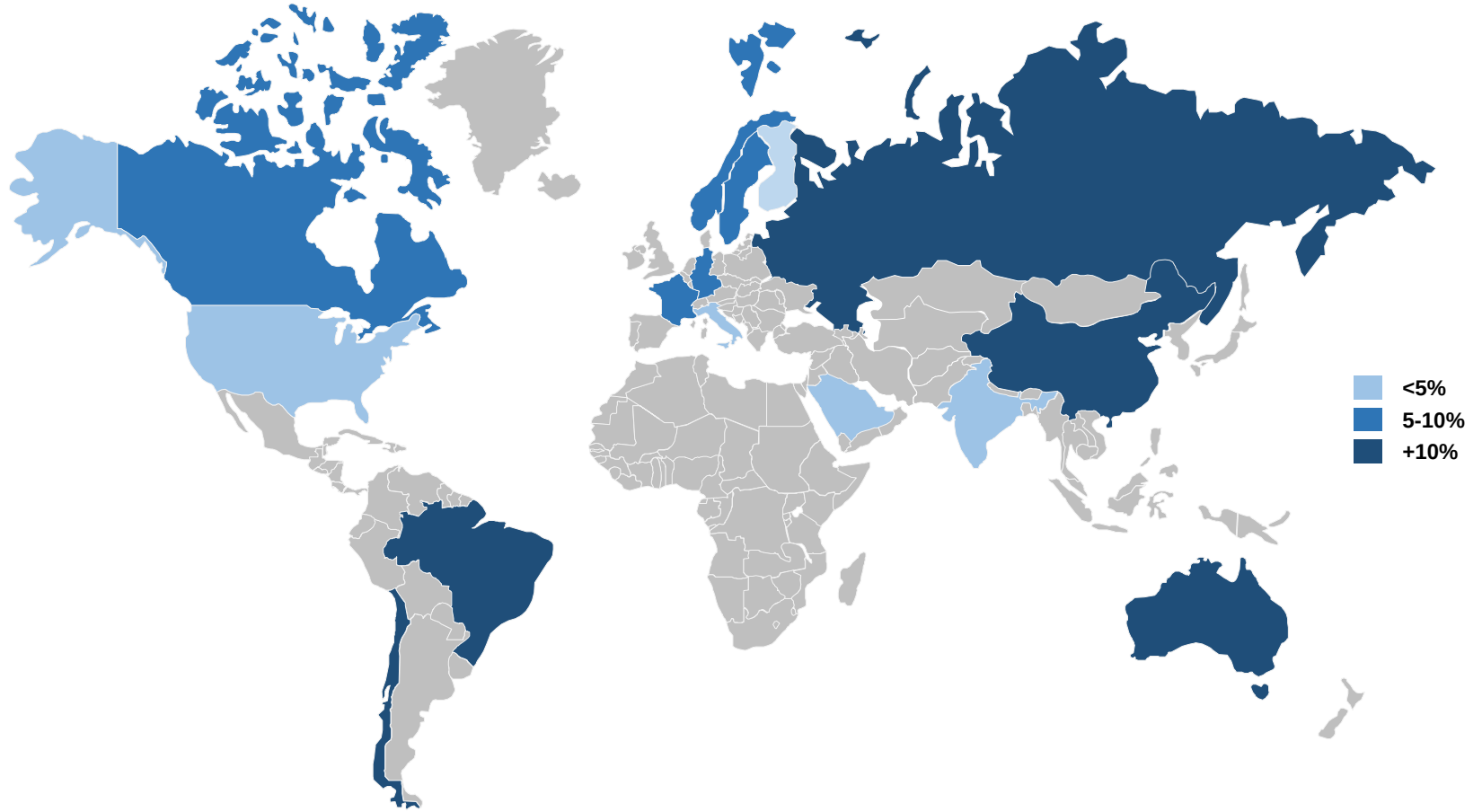
Chart 2

- Chileans prefer to shop in stores, as Chinese
- Brazilian market is 'stuck in the middle' with about half of the population preferring to shop online (slight preference for online)
- Australian and Russians are keen on online shopping with more than 70% of their population embracing this way of buying clothes

Clothing retailer

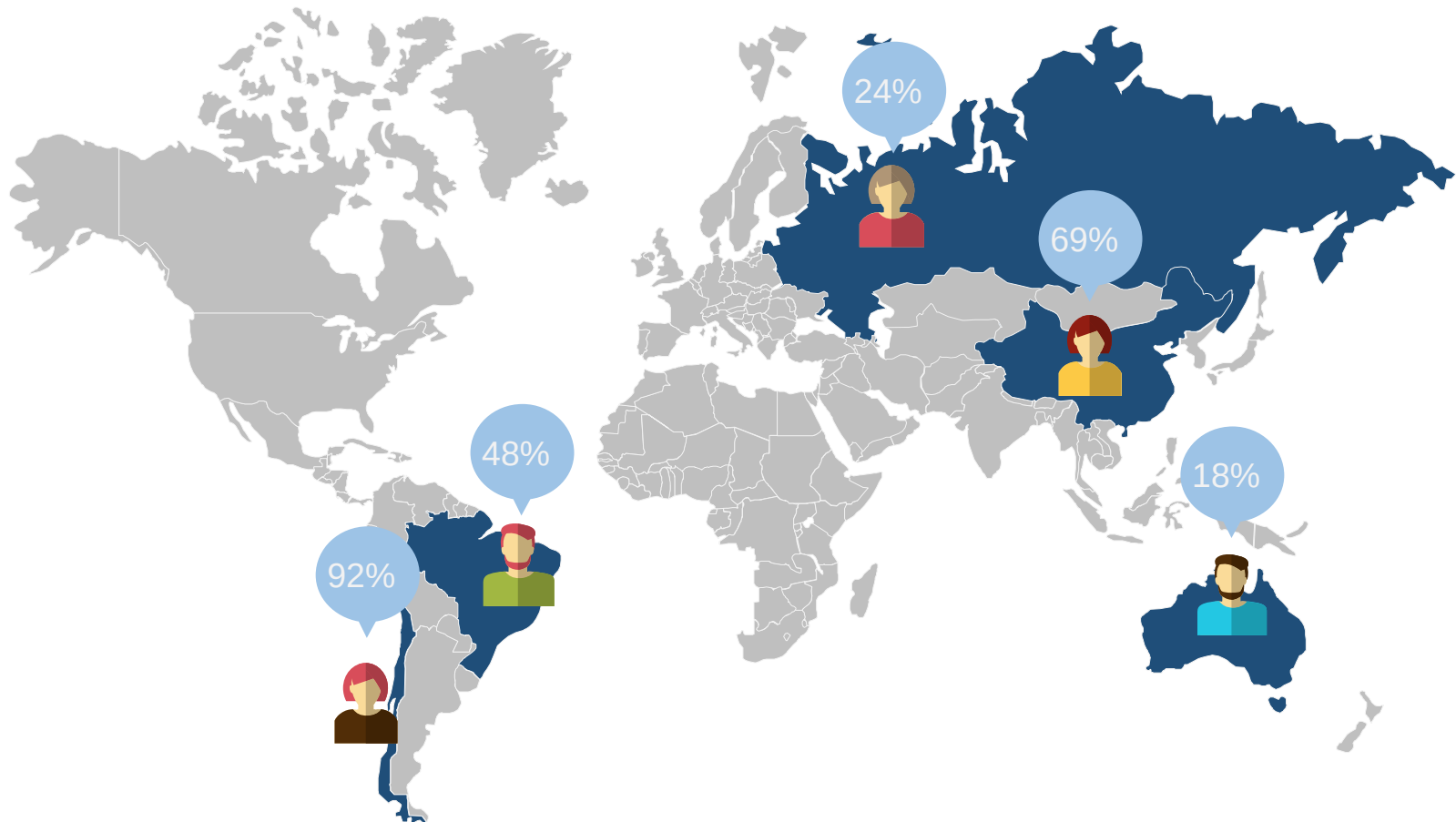
World's largest markets – Clothing Industry

In terms of market growth (in %), 2018 data



Clothing retailer

Customer preference in ways of shopping for clothes
% of customer preferring to shop in retail stores, 2018 data



Clothing retailer

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Market sizing

Your client is leaning towards Australia for expansion as this market is growing, resembles the Canadian market and would require less upfront investment (based on management's first analysis). Before expanding, management would an assessment on the size of Australian's market. **Could you size the market for clothes in Australia?**

Suggested approach

Top-down approach

- Aussie population: 24M
- Life expectancy: 80y
- Age distributed evenly (in 4 groups – 6M/group)
- Men 50% vs. women 50% of population
- Average spent on clothes
 - Men (12M):
 - 0-20: \$25/mo. * 12 mo. * 3M people
 - 20-40: \$50/mo. * 12 mo. * 3M people
 - 40-60: \$100/mo. * 12 mo. * 3M people
 - 60-80: \$50/mo. * 12 mo. * 3M people
 - Women (12M) Spending 3x as much as men
 - 0-20: \$75/mo. * 12 mo. * 3M people
 - 20-40: \$150/mo. * 12 mo. * 3M people
 - 40-60: \$300/mo. * 12 mo. * 3M people
 - 60-80: \$50/mo. * 12 mo. * 3M people

\$~29B/yr.

Spent in Clothing in Australian market

Most important is not the final answer but candidate's approach to find answer

Bonus question

Recent study shows that they could capture a 15% market share. What would be their total revenue under this scenario?

Answer

\$~4.35B

Conclusion

Benedict's management team would like to know if they should move ahead with their international venture. The managers are meeting next week to go define what to do next. Based on your analysis, **what is the best move for the company in terms of international expansion?**

Suggested Answer

- Enter Australian market through an online store
 - 3 reasons
 - Similarity with Canadian market in terms of size and demand
 - Thriving market with a +10% annual growth
 - Low capital required in order to enter market through online store
- Nevertheless, the company has to be mindful of risks associated with entering a new market. For example:
 - High competition from low-cost brands already present on the market

Trainer's guide

TO PROMPT

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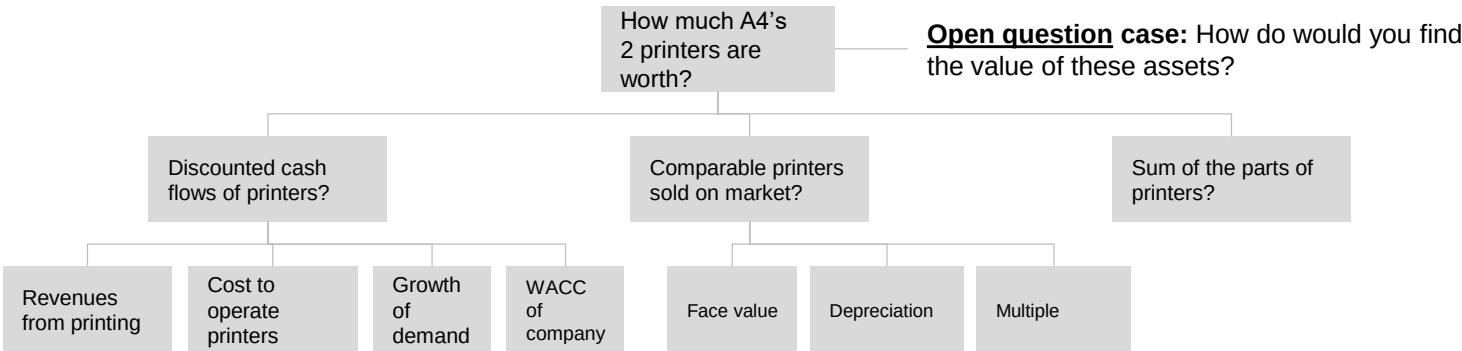
Opening statement

Your team has been mandated to advise a printing company called A4. A4 is specialized in printing encyclopedias. The market for encyclopedias is pretty standardized and mature. Three types of printers are used to print the end product: custom, standard and industrial. Printers' output is similar regardless of type, but 'custom' printers allow for greater flexibility on end product. A4's sole owner has seen an incalculable number of pages being printed in his lifetime and now retirement is looming. **In this context, he would like to sell his 2 standard printers and has asked for the help of your team to value these assets.**

Further information

- Only 1 kind product printed by the company: encyclopedias
- Value chain not integrated (i.e., not part of a publishing house)
- Encyclopedias are standardized on this market and there is no difference between offering from printing companies
- Clients' printers are fully depreciated
- The 3 kinds of printers can only print 1 type of encyclopedia
- Printers only differ on capacity and cost (see below)
- Printing capacity:
 - Custom: 200 pages/week
 - Standard: 400 pages/week
 - Industrial: 600 pages/week
- Cost to operate
 - Custom: \$1 000/encyclopedia
 - Standard: \$500/encyclopedia
 - Industrial: \$250/encyclopedia
- A month has 4 weeks
- Number of printers on market
 - Custom: 100
 - Standard: 200
 - Industrial: 400
- 1 encyclopedia has 1 200 pages
- Costs of operations: paper, ink and electricity to run machines
- Demand: 1 000 encyclopedias/month
- No information on price charged to clients
- Market is decreasing 10% a yr. because of paperless trend
- Client would like to use discounted cash flows method to value his assets
- WACC of company (Weighted Average Cost of Capital) = 5%

Suggested structure



Printing Co

Trainer's guide

TO PROMPT

TO PROVIDE IF REQUESTED

SUGGESTED SOLUTIONS

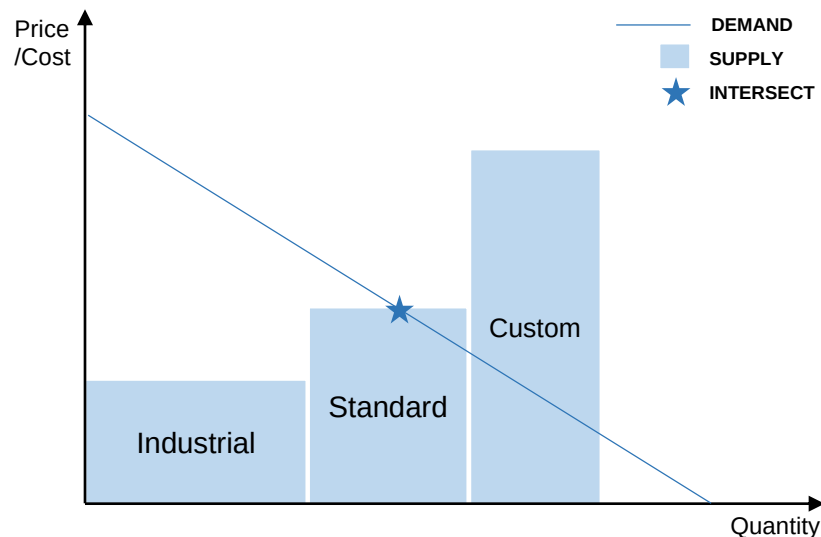
Valuation

Calculate the value of the printers according to the method privileged by the client.

Further information

- No information is available in terms of price/encyclopedia charge to clients
- The product offered by the company is a commodity
- Most efficient players will win on this market (i.e., players with the lowest marginal cost will produce first until capacity is reached, then 2nd most efficient will operate, and so on and so forth)
- To solve this question, trainee has to use economics theories about competitive market. See below for a graphical representation of the answer.

Suggested Answer



- Trainee has to understand that most efficient suppliers will produce first (i.e., industrial printers)
- Only part of 'standard' printers will produce on this market (demand < capacity)
- Custom printers are not operating in this market since not efficient enough
- Since the market is competitive, there is uncertainty about which 'standard' printers will produce
- Therefore, which 'standard' printers is producing will be defined by lowest offered price (i.e., because of competitive dynamics marginal cost will be = price)
- Hence, $\text{Marginal Revenue} - \text{Marginal cost} / (\text{WACC} - \text{Growth}) = 500 - 500 / 5\% - (10\%) = 0$
- Both printers have no value when using DCF method

Printing Co

Trainer's guide

TO PROMPT

TO PROVIDE IF REQUESTED

SUGGESTED SOLUTIONS

Creativity question

Interesting analysis. **Do you think our client could find a way to differentiate his offer and generate other revenue streams from printers? If so, how?**

Suggested answer

- Offer better financing solutions to clients
- Print on new type of paper (e.g., high quality paper)
- Reduce time to deliver by placing facilities closer to end-client
- Reduce shipping costs

Most important is to define creative solutions to the clients situations. Additionally, the trainee has to try to answer this question in a structured way

Conclusion

To conclude, based on what you know, what is the value of the 2 printers?

Suggested answer

- The printers valued is 0 for 3 reasons
 - Discounted cash flow = 0
 - Declining market at a rate of -10%/year in terms of revenues
 - Low potential to differentiate from competitors with current assets
- However, there might be residual value of assets by selling parts of printers part by part
- If I had more time, I would have analyzed how to value parts of printers

Trainer's guide

- TO PROMPT
- TO PROVIDE IF REQUESTED
- SUGGESTED SOLUTIONS

Opening statement

Gretzky State University, GSU, is a graduate school operating in sunny New Mexico. Its beautiful campus attracts students from all over the United States. Today, the school's revenues are growing but at a slower pace than it did before. In the eyes of GSU's board, this can be explained by the saturation of the American market in terms of grad school offering. Consequently, Gretzky State is looking to diversify its activities and would like to capture new income opportunities. International expansion is the favored option by the board. In this context, the Dean of the institution has just approached Chinese University, CU, a school that is in the process of creating a new campus in China and that is actively looking for a business partner for this project. In this context, CU has offered a joint venture deal to the Dean of GSU where both players would have equal equity in the project. **Do you think GSU should enter this partnership?**

Further information

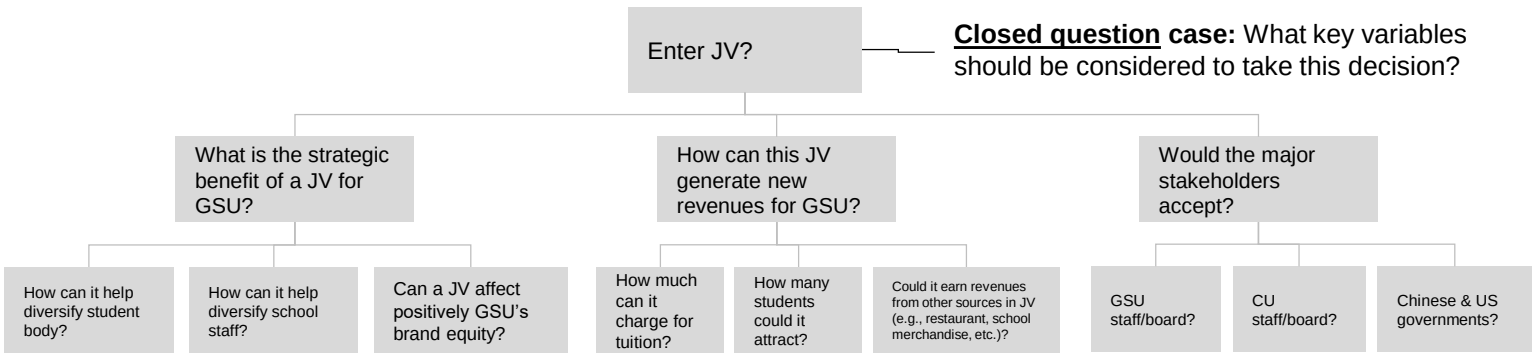
GSU's Profile

- Entering a joint venture (JV) is new for GSU
- The board supports the joint venture but most of school's staff don't
- Strong brand equity as GSU was founded in 1850 and is always highly ranked in national rankings
- Offers only Masters of Art degrees (1 campus)
- GSU has increasing concerns about the homogeneity of its student body and staff; would like to increase diversity to improve ranking internationally
- Average tuition: \$100K/yr.
- 2 000 students
- Goal of JV: Diversify student body/staff + generate new revenue streams (no concerns about costs)

CU's Profile

- Currently operates in 10 different markets (10 campuses) and has experience in opening new locations
- Offers Law and Business degrees
- Brand image of school is still to develop
- Average tuition: \$83K/yr.
- 9 000 students
- The current school staff is sceptic about the JV
- Goal of JV: Gain brand equity by capitalizing on GSU's branding

Suggested structure



Charts
questions

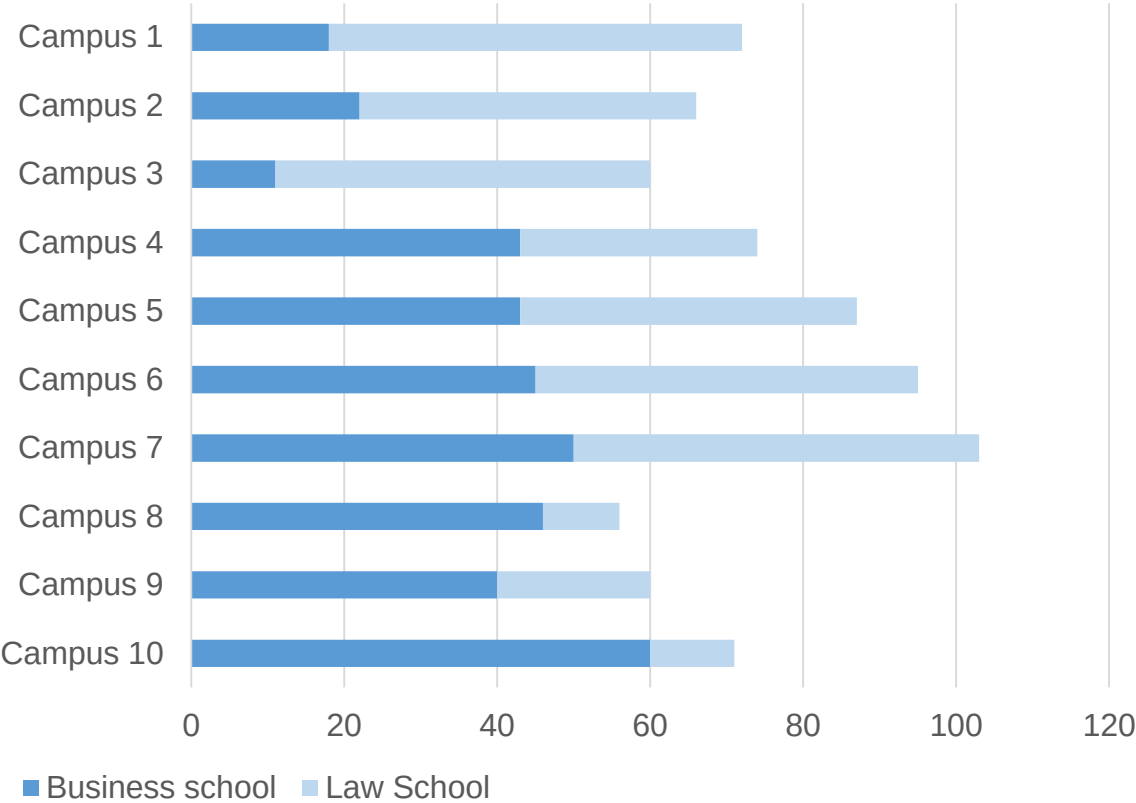
In the due diligence process, GSU has managed to obtain information on CU's current revenues and student body profile. **What can you conclude for analyzing these charts?** Prompt 2 charts to trainee.

Suggested
Insights

- 9 000 students spread across 10 campuses; the school is larger in terms of number of students than GSU
- Low income students have an incentive to graduate (\$10K check upon graduation)
- CU has also an incentive to graduate low income students (\$17K upon graduation)
- No correlation between financial background and probability of graduating - could be related to the government's program
- The candidate should be able to identify 3 markets:
 - 1 concentrated in Business school with little Law school revenues (campuses 8, 9 and 10)
 - 1 concentrated in Law school with little business revenues (campuses 1, 2 and 3)
 - 1 equally diversified generating the most revenues (Campuses 5, 6 and 7)
- The diversified market offers the highest revenues since it attracts mostly people from low income background (one potential explanation)
- Offering Business and Law degrees in China would increase diversity of student body for GSU
- Offering an equal mix of Business and Law degrees would be a way to maximize revenues from joint venture

School

Revenues generated by CU
\$M by campus, 2018 data



Financial background of CU's
graduate students
In thousands, 2018 data



Note 1: Chinese government has established a 'bonus upon completion' program in 2018. Every student from a low income background that completes a masters degree receives a \$10K check from the government while the graduate school receives \$17K.

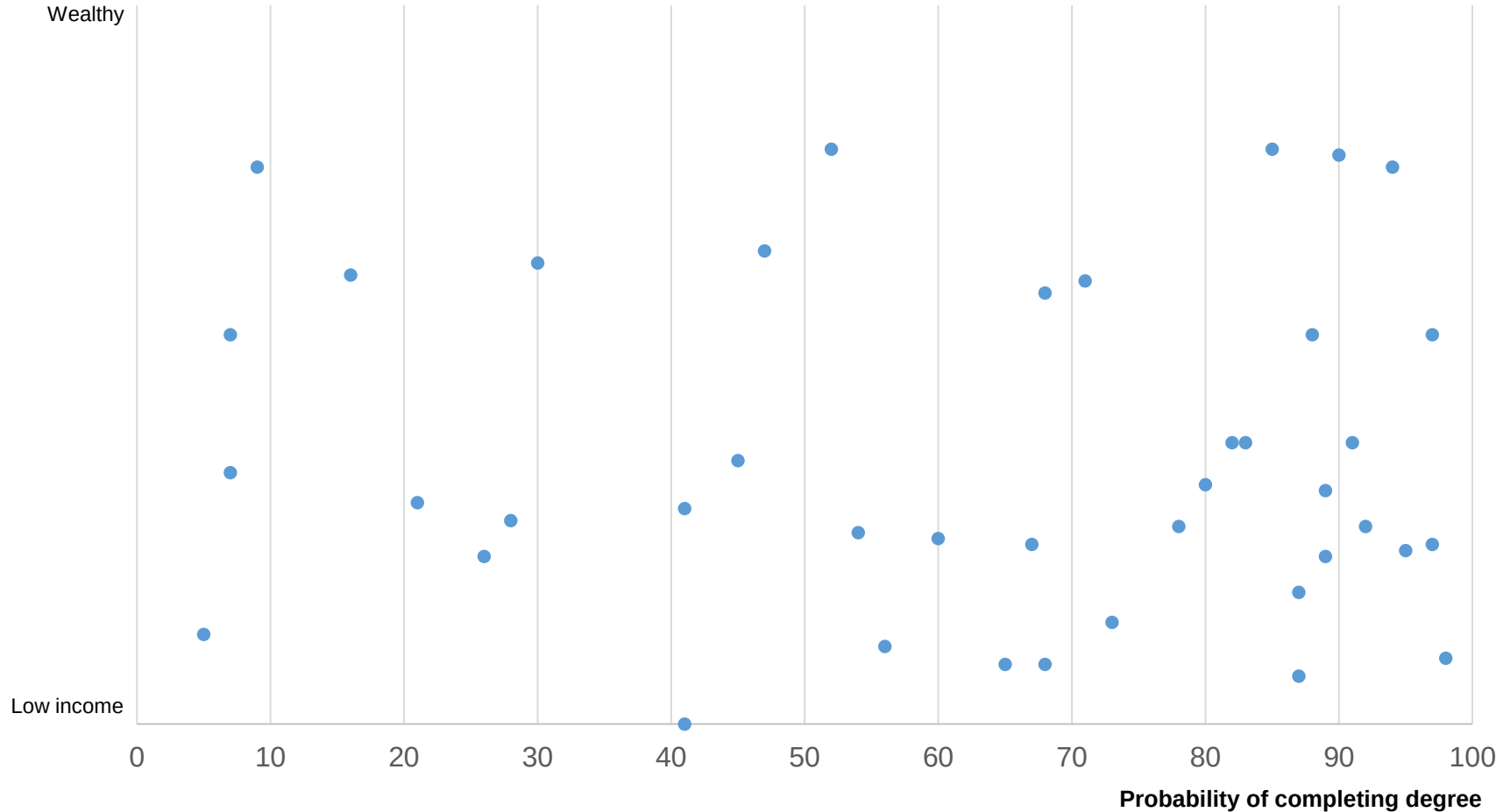
School

Financial background vs. probability of completing degree in CU's grad school

Correlation, 2018 data

Financial background

Wealthy



Creativity
question

If we put the joint venture idea aside for a minute, **what could be other revenue uplift opportunities for GSU?**

Suggested
answers

- Higher tuition fees
- New programs, same campus
- New campus, same program
- New campus, new programs
- Online courses
- Summer school
- Leadership seminars for professionals

Most important is candidate's creativity to address revenue issue of Gretzky State

Bonus points for answers that would also tackle/consider need for diversity among student body

Conclusion

What do you think GSU should do?

Suggested
answered

- Enter JV
 - New revenue streams: \$83K/student and \$100K/low income graduating student (83K + 17K of gov. program)
 - Diversity of student body: Enlarged student body in terms of financial background and nationality (no information available on school staff)
 - New programs offering: Business & Law
- Risks
 - Would staff of both schools accept?
 - Would CU's brand image affect negatively GSU's brand equity?
 - Are US and Chinese governments onboard with this collaboration?
 - Operating model might be complex with two schools managing this new campus

How would you approach these business situations?

Opening Drills

To prompt to trainee

Advice: Don't focus on answering the question. Rather, define how to approach the situation best

1. You are mandated by the Saudi government. They would like to diversify their economy by moving away from the oil industry and be fully diversified by 2030. How should they approach this issue?
2. Apple has hired your team to define a strategy going forward with the iPad. Sales are plummeting and Apple's market share is shrinking. Should they continue developing hardware updates for the iPad?
3. The French government wants to foster entrepreneurship among its population. What are the options it can explore and how to choose among them?
4. A large newspaper, the LA Times, has hired you to clarify how AI could threaten its business model and if it should embrace the technology to help journalists in article writing. How would you approach this issue?
5. The family owners of housing rental business in North Carolina are worried about the increasing number of hurricanes in the region. Facing this threat, should they close business and move it elsewhere?
6. FIFA is wondering how to approach the future of World Cups to ensure these events be profitable ventures for hosting countries. How would you study this issue?
7. Your client is a large distribution company that would like to know if they should invest in manufacturing capability to reduce time to market. What should be analyzed in order to take this decision?
8. Airbnb is facing higher and higher regulations in most of their market and would like to understand why this is happening and how to tackle this issue best?
9. The partners of a major consultancy wonder if they should raise the partnership buy-in. What should they consider if order to take a decision?
10. Should Nike acquire Adidas?

Trainer's corner

Did the candidate...

- ☐ Tailor structure to the client's question?
- ☐ Has at least 2 sublayers in structure?
- ☐ Develop a structure in 30-45 sec.?
- ☐ Frame structure as questions?
- ☐ Develop a MECE structure?
- ☐ Executive summarize structure before deep diving?
- ☐ Present structure in ~2 min.?

How would you approach these business situations?

Opening Drills

To prompt to trainee

Advice: Don't focus on answering the question. Rather, define how to approach the situation best

11. Perrier, a mineral water producer, is witnessing new competition from smaller players on its market. It is now wondering if it should acquire its largest competitor to consolidate its position in the industry. Should they do it?
12. A major publishing house is contemplating crowdsourcing text reviews (instead of performing the activity in-house). How would you analyze this issue?
13. A business school in the US is wondering how it could attract more international students to its MBA program. How should the board of directors approach this business situation?
14. Patagonia, a outdoors clothing manufacturer, is thinking about going public (IPO) to raise fund for a new sustainable initiative. How would you address this question?
15. TO, a large PE firm, is pondering how to best select its next investment. Its partners have hired you to help them solve the issue.
16. Hawaii, a US state consisting of 8 islands in the Pacific ocean, is considering privatizing its beaches and would like to know if they should do it.
17. A large manufacturer of pools is seeing its sales declined. Its management would like to pump-up the sales team. What would you analyze in order to solve the issue?
18. Your client is a bar operator in the beautiful city of Paris and is wondering if they should open a new spot in the small city of Jouy-en-Josas. How should they address this question?

Trainer's corner

Did the candidate...

- ☐ Tailor structure to the client's question?
- ☐ Has at least 2 sublayers in structure?
- ☐ Develop a structure in 30-45 sec.?
- ☐ Frame structure as questions?
- ☐ Develop a MECE structure?
- ☐ Executive summarize structure before deep diving?
- ☐ Present structure in ~2 min.?

Can you define an approach to find a ‘good enough’ estimate of the market?

Market Sizing Drills

To prompt to trainee

Advice: Don't focus on answering the question. Rather, focus on approach to defining a 'good enough' estimate

Try to define an approach to reach ‘good enough’ estimates of these markets

1. Total value of the chewing gum market in the US
2. Number of smartphones sold in Europe last year
3. Demand for commercial flights from Singapore to Delhi per year
4. Number of commercial printers sold in the US last year
5. Total yearly revenue of management consulting services in Europe
6. Number of PlayStation gaming consoles sold in Europe last year
7. Number of Bloomberg terminals sold in London last year
8. Total demand in terms of revenues for Paris-Rome flights
9. Number of credit card transactions made in Thailand last year
10. Total market value of dog grooming industry in the US

Trainer's corner

Did the candidate...

- ☐ Present approach before asking for figures?
- ☐ Identify all key factors to size the market?
- ☐ Present ‘reasonable’ assumptions?

Advice: Move away from ‘pure’ sizing questions (e.g., number of tennis balls that can fit in a Boeing 737) as they don't assess business judgment

How efficiently can you solve math questions?

Math Drills

To prompt to trainee

1. If \$2M of revenue represents 30% market share, what is the total market value?
2. Our marginal cost is \$30 and we are earning a revenue of \$70 per product sold. If our fixed cost is \$700 000, how many units do we have to sell in order to break even?
3. Today, our market share is 85%. If it decreases by 15% next year, what will be our new share of the total market?
4. I earn a total revenue of \$7 600 by selling 13 units. What's my revenue per unit?
5. I sell 15 units at a price of \$14.78/unit. What's my total revenue?
6. My total profits is \$876B and will be divided by 20 000 next year. What's my expected revenue for next year?
7. If my car has a tank capacity of 40 liters and offers a performance of 65km/liter, how many kilometers can I expect to do with a full tank?

Advice: Change figures to adjust difficulty of math drills

Trainer's corner

Did the candidate...

- ☐ Walk the trainer through calculations' rationale?
- ☐ Do calculations properly?
- ☐ Ask before rounding figures?
- ☐ Do calculations fast enough?
- ☐ Stay engaging?
- ☐ Sanity check results?

Reflect upon, prepare and rehearse fit questions

Advice: Practice answering fit questions in less than 2 minutes for greater impact!

★ KEY IN CONSULTING INTERVIEWS

Category	#	Question
(A) General	1	Tell me about yourself / Discuss your resume
	2	What are your strengths?
	3	What are your weaknesses? What are three things you former manager would like you to improve upon?
	4	Where do you see yourself in five years? Ten years?
	5	Why are you interested in working for us?
	6	4 words to describe yourself
	7	What are you salary requirements?
(B) Consulting ★	8	Tell me about a time you had to convince someone
	9	Provide an example of a time that you shown leadership skills?
	10	Tell me about a time you shown entrepreneurial skills
	11	What would you like the most and least about working in the consulting industry?
	12	Tell me more about the kind of consulting you would like to do (e.g., Strategy, Ops, etc.)
	13	Tell me about a personal challenge you overcame
	14	Tell me about the accomplishment you are most proud of
	15	Who are our competitors? How are we different?
	16	Why you'd be a good strategist?
	17	What are some of your leadership experiences?
	18	Provide example of work that you done in strategy consulting (or close to)
(C) Preferences	19	Are you willing to travel?
	20	What's your availability?
	21	Would you work holidays/weekends?
	22	Would you work 60+ hours a week?

Reflect upon, prepare and rehearse fit questions

Advice: Practice answering fit questions in less than 2 minutes for greater impact!

★ KEY IN CONSULTING INTERVIEWS

Category	#	Question
(D) Situations ★	23	Describe a time when you had to analyze data to create a solution to a problem
	24	Give me an example of a time when you had to build a relationship with someone. How was the person related to your goals and expected outcomes? What did you do to initiate the relationship building?
	25	Tell me an example of where you have used analysis to deliver project improvements
	26	Tell me how you handled a difficult situation
	27	Tell me about a time when you received a piece of developmental feedback. How did you react and what did you do with this feedback?
	28	How would you deal with an angry client?
	29	Give me an example of a situation when you had to cope with great pressure or deliver something on a very tight deadline
	30	Tell me about a time when you have been faced with dishonest behaviour? What was the situation and what did you do?
	31	Describe a time where you had to confront a difficult interpersonal situation at work. How did you handle it?
	32	Tell me about a time when your communication skills made a difference to a situation
	33	Give me a time when you went above and beyond the requirements for a project
	34	Can you give me an example of where you collaborated across teams on a project?
	35	Describe a time where you had to depend on others to complete a task
	36	Tell me about a time when you worked with a difficult team member. What was the situation and how did you handle it?
	37	Give a specific example of when you had to handle priorities. How did you meet all of your responsibilities?
	38	Tell me about a time you delivered exceptional customer service
	39	Give me an example of where you delivered a change that improved the client experience?
(E) Last job	40	Why did you leave your past job?
	41	Why was there a gap in your employment between 2015 and 2016?
	42	If I called your boss right now and asked him what you could improve on?
	43	Explain clearly your role at your last job
	44	Why are you looking for a new job?

Reflect upon, prepare and rehearse fit questions

Advice: Practice answering fit questions in less than 2 minutes for greater impact!

Category	#	Question
(F) Personal	45	What motivates you?
	46	Why should we hire you?
	47	What is your dream job?
	48	What is your biggest failure?
	49	What makes you uncomfortable?
	50	What are your co-worker pet peeves?
	51	What are your hobbies?
	52	What is your favorite website?
	53	What are your career goals?
	54	What gets you up in the morning?
	55	What was the last book you read for fun?
	56	What questions do you have for me?

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